

TE Curve Verification Pipeline Familywise ONNX Report - wave4_3_mixture_density_k2

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k2/2026-07-15-09-53-26__track2_wave4_3_mixture_density_k2_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k2/2026-07-15-09-53-26__track2_wave4_3_mixture_density_k2_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k2/2026-07-15-09-53-26__track2_wave4_3_mixture_density_k2_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_3_mixture_density_k2/2026-07-15-09-53-26__track2_wave4_3_mixture_density_k2_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_3_mixture_density_k2` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_3_mixture_density_k2_fw_simplified_setpoints</code>	<code>2026-07-14-23-46-20__te_wave4_3_mixture_density_k2_fw__simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_3_mixture_density_k2_bw_simplified_setpoints</code>	<code>2026-07-15-00-37-24__te_wave4_3_mixture_density_k2_bw__simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_3_mixture_density_k2_global_simplified_setpoints</code>	<code>2026-07-14-22-56-56__te_wave4_3_mixture_density_k2_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/forward/2026-07-14-23-46-20__te_wave4_3_mixture_density_k2_fw__simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/forward/2026-07-14-23-46-20__te_wave4_3_mixture_density_k2_fw__simplified_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=258-val_mae=0.00346706.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/backward/2026-07-15-00-37-24__te_wave4_3_mixture_density_k2_bw__simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/backward/2026-07-15-00-37-24__te_wave4_3_mixture_density_k2_bw__simplified_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=185-val_mae=0.00360751.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/global/2026-07-14-22-56-56__te_wave4_3_mixture_density_k2_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_3_mixture_density_k2/global/2026-07-14-22-56-56__te_wave4_3_mixture_density_k2_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=237-val_mae=0.00346790.ckpt</code>

Aggregate Metrics

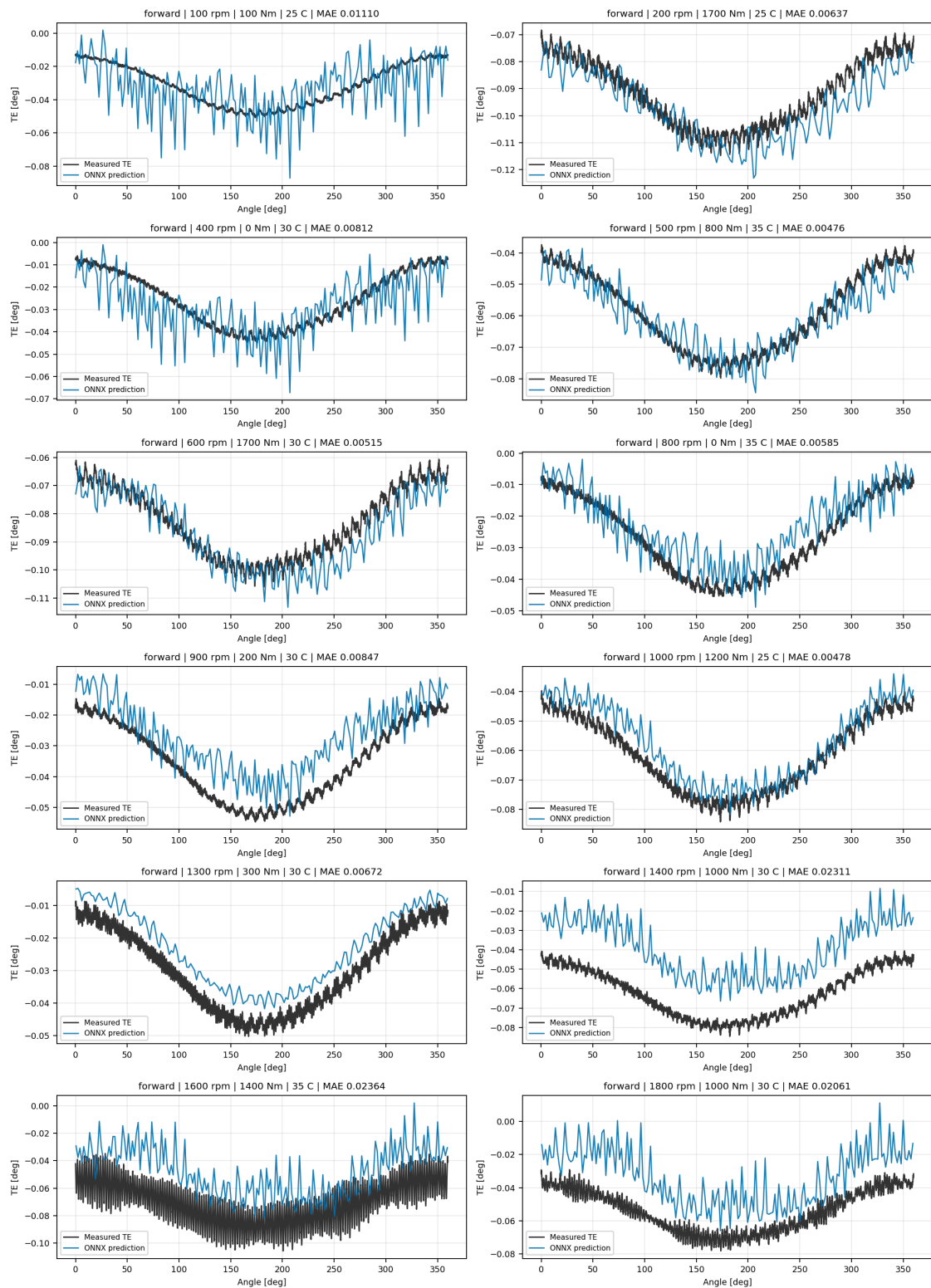
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.010679	0.012442	24.812	54.414
backward	97	0.011148	0.013701	26.465	42.420
global	194	0.022125	0.027346	52.474	81.762

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.004585	0.009190	0.005742	0.016011
backward	-0.005816	0.007644	0.008396	0.029506
global	0.005751	0.011646	0.019236	0.097188

Forward 12-Curve Page

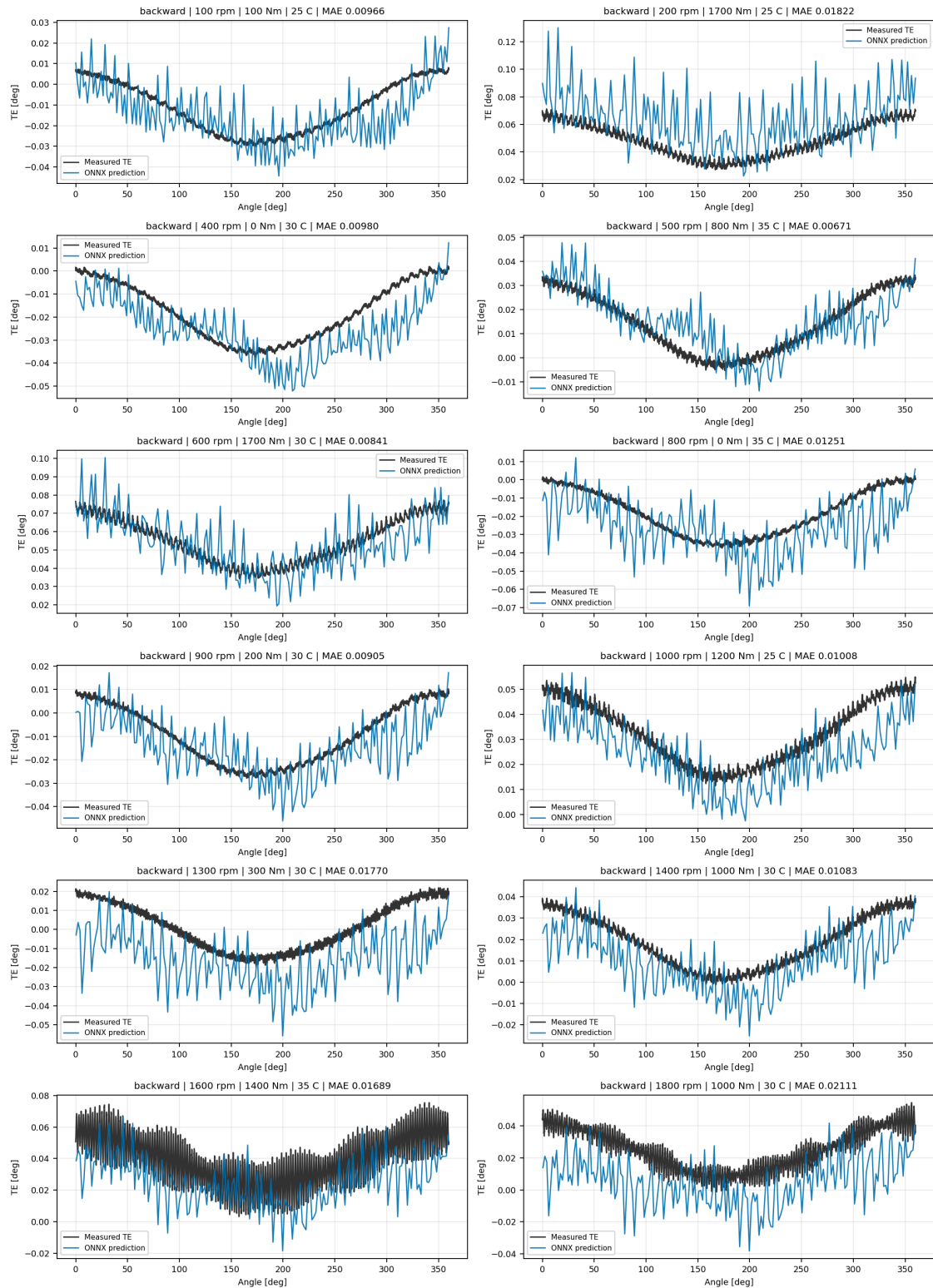
Simplified Dataset + Setpoints | wave4_3_mixture_density_k2 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

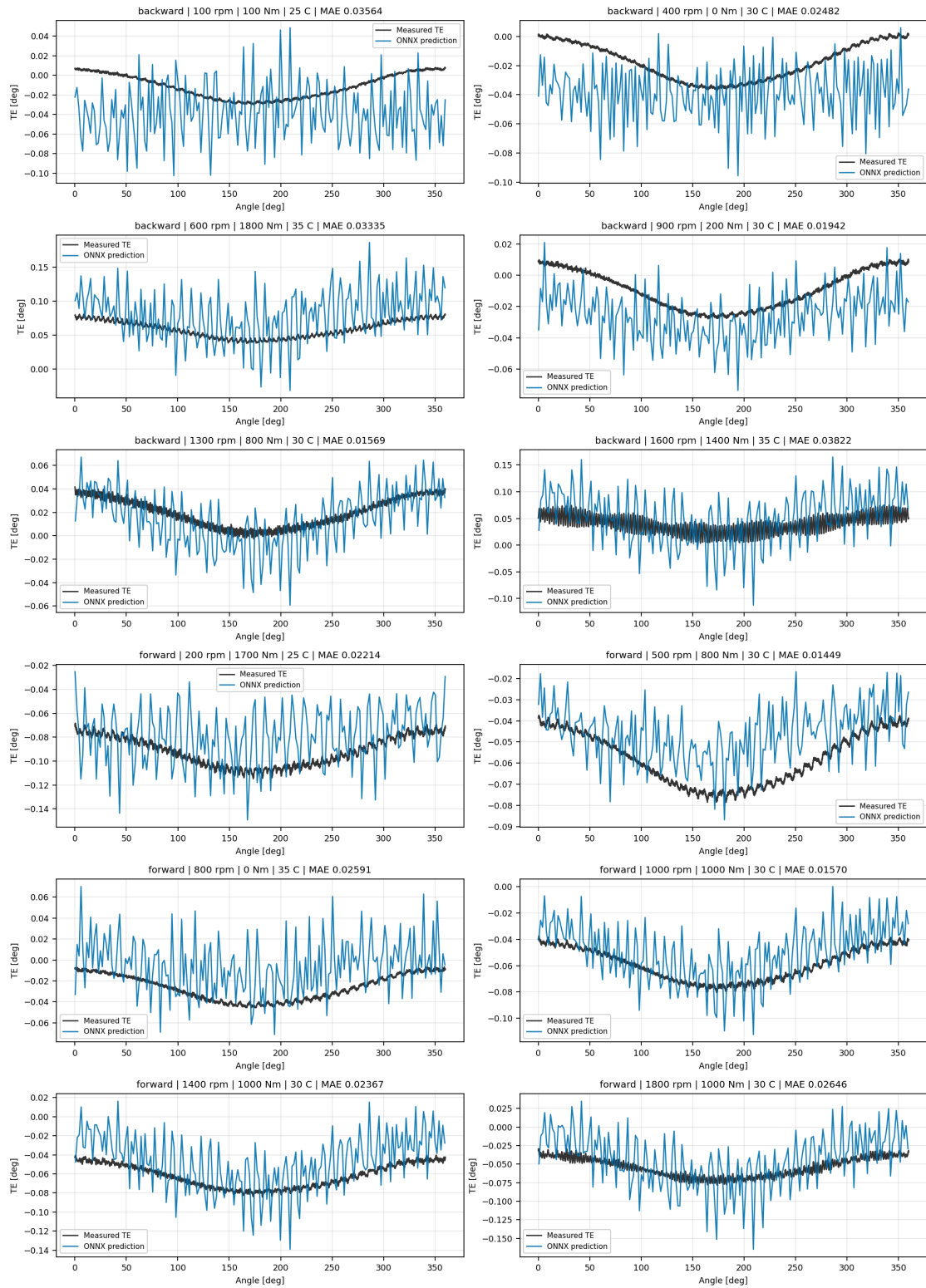
Simplified Dataset + Setpoints | wave4_3_mixture_density_k2 | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_3_mixture_density_k2 | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_3_mixture_density_k2 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_3_mixture_density_k2_fw_polished_setpoints	2026-07-15-02-19-32_te_wave4_3_mixture_density_k2_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_3_mixture_density_k2_bw_polished_setpoints	2026-07-15-02-51-55_te_wave4_3_mixture_density_k2_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_3_mixture_density_k2_global_polished_setpoints	2026-07-15-01-42-15_te_wave4_3_mixture_density_k2_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/forward/2026-07-15-02-19-32_te_wave4_3_mixture_density_k2_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/forward/2026-07-15-02-19-32_te_wave4_3_mixture_density_k2_fw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=101-val_mae=0.00184963.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/backward/2026-07-15-02-51-55_te_wave4_3_mixture_density_k2_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/backward/2026-07-15-02-51-55_te_wave4_3_mixture_density_k2_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=173-val_mae=0.00181733.ckpt
global	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/global/2026-07-15-01-42-15_te_wave4_3_mixture_density_k2_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_3_mixture_density_k2/global/2026-07-15-01-42-15_te_wave4_3_mixture_density_k2_global_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=122-val_mae=0.00186311.ckpt

Aggregate Metrics

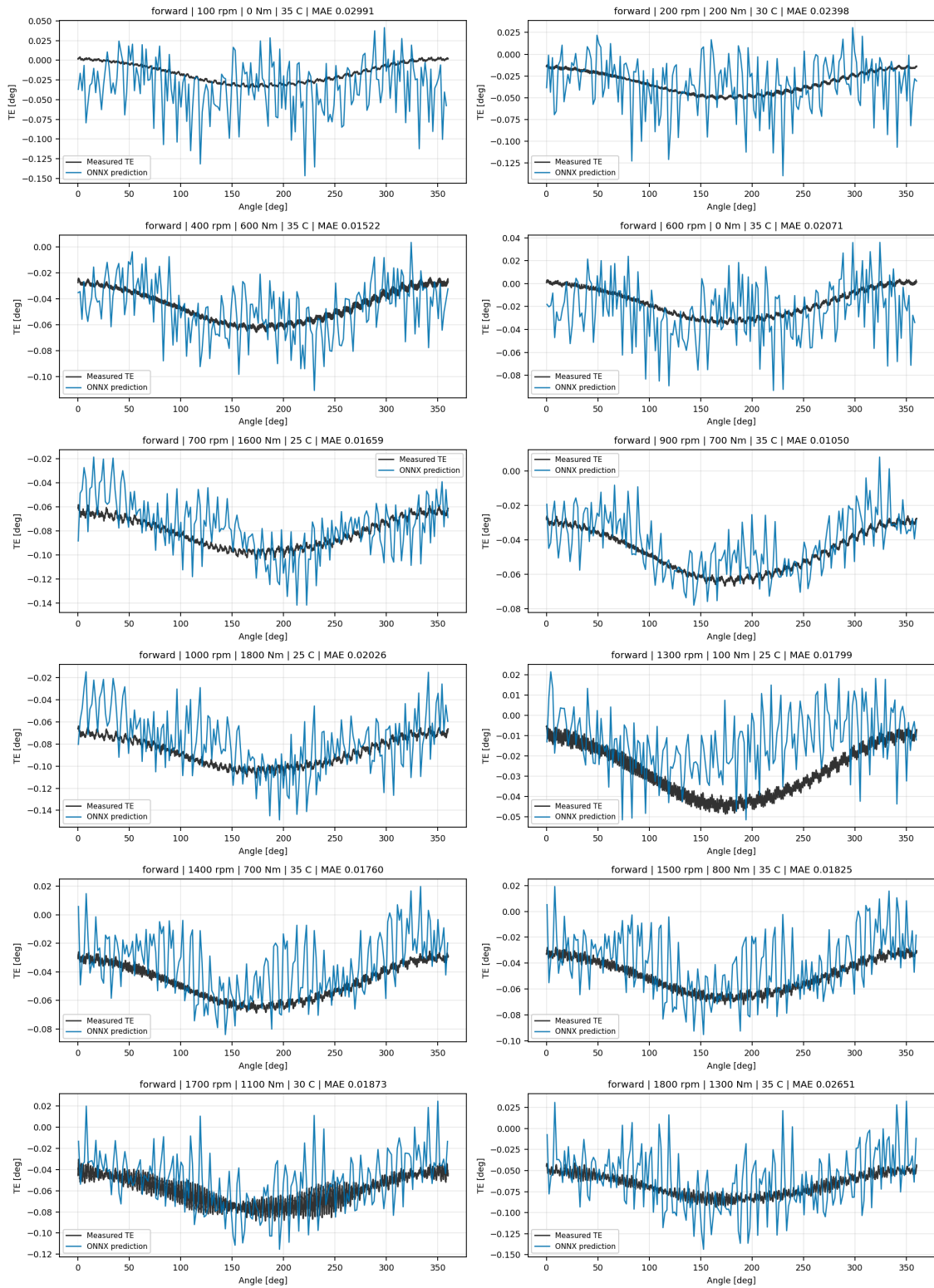
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.017215	0.021524	40.412	60.359
backward	94	0.013118	0.015721	30.408	60.543
global	194	0.019799	0.024430	46.824	72.498

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.004753	0.006789	0.015895	0.071707
backward	-0.007703	0.009631	0.008996	0.017015
global	-0.000462	0.010660	0.016442	0.074638

Forward 12-Curve Page

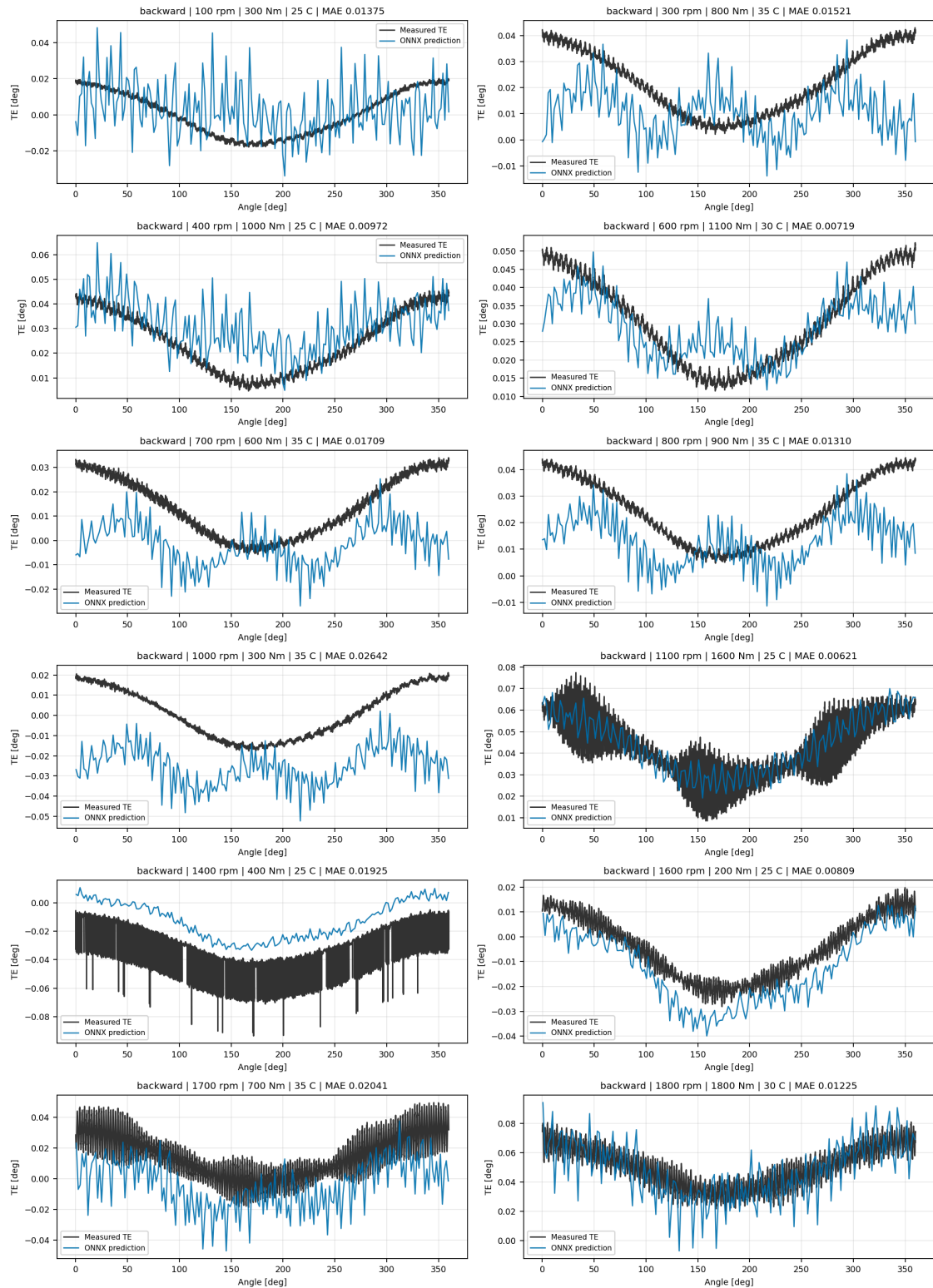
Polished Dataset + Setpoints | wave4_3_mixture_density_k2 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

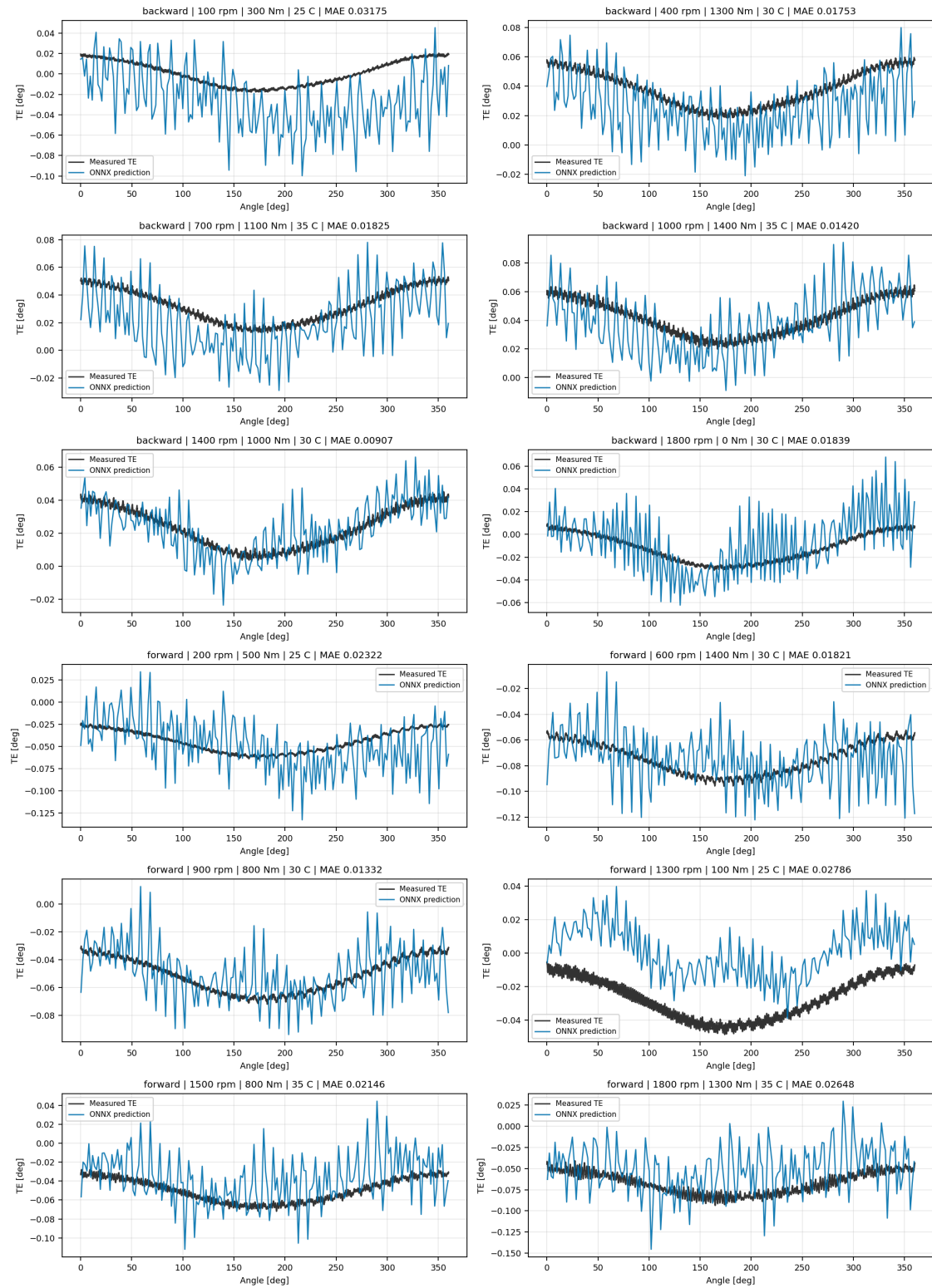
Polished Dataset + Setpoints | wave4_3_mixture_density_k2 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_3_mixture_density_k2 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_3_mixture_density_k2 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_3_mixture_density_k2_fw_polished_actual_values	2026-07-15-05-01-50_te_wave4_3_mixture_density_k2_fw_polished_actual_values	polished_point_v1
backward	te_wave4_3_mixture_density_k2_bw_polished_actual_values	2026-07-15-06-02-16_te_wave4_3_mixture_density_k2_bw_polished_actual_values	polished_point_v1
global	te_wave4_3_mixture_density_k2_global_polished_actual_values	2026-07-15-04-04-29_te_wave4_3_mixture_density_k2_global_polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/forward/2026-07-15-05-01-50_te_wave4_3_mixture_density_k2_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/forward/2026-07-15-05-01-50_te_wave4_3_mixture_density_k2_fw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=256-val_mae=0.00172453.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/backward/2026-07-15-06-02-16_te_wave4_3_mixture_density_k2_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/backward/2026-07-15-06-02-16_te_wave4_3_mixture_density_k2_bw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=129-val_mae=0.00180093.ckpt
global	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/global/2026-07-15-04-04-29_te_wave4_3_mixture_density_k2_global_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_3_mixture_density_k2/global/2026-07-15-04-04-29_te_wave4_3_mixture_density_k2_global_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=252-val_mae=0.00175520.ckpt

Aggregate Metrics

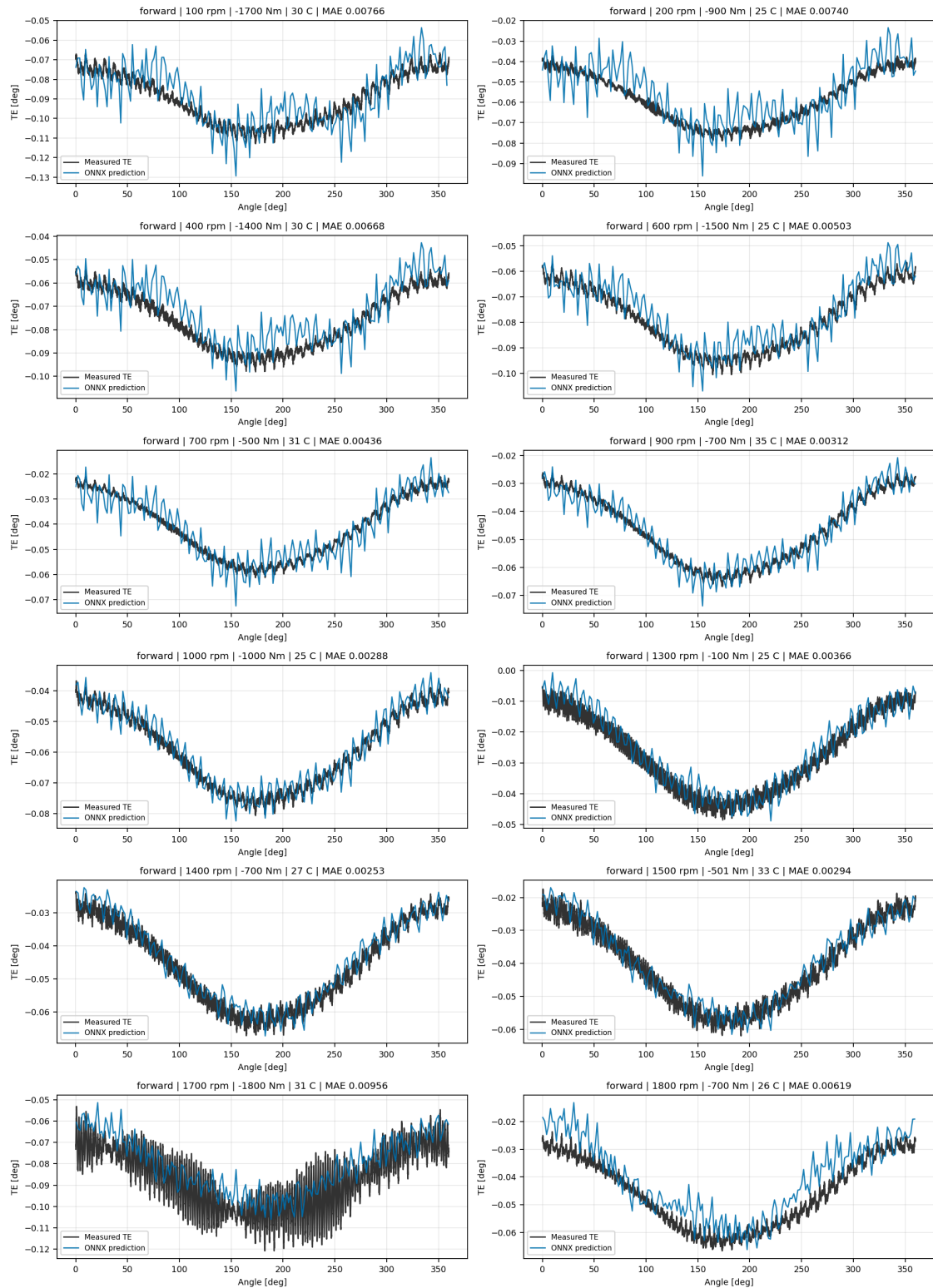
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.005174	0.006329	12.160	21.195
backward	94	0.019883	0.024875	46.089	64.564
global	194	0.014640	0.017919	33.940	57.444

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.001978	0.002181	0.004645	0.014298
backward	-0.001380	0.005265	0.019178	0.087269
global	0.001971	0.009325	0.011718	0.044567

Forward 12-Curve Page

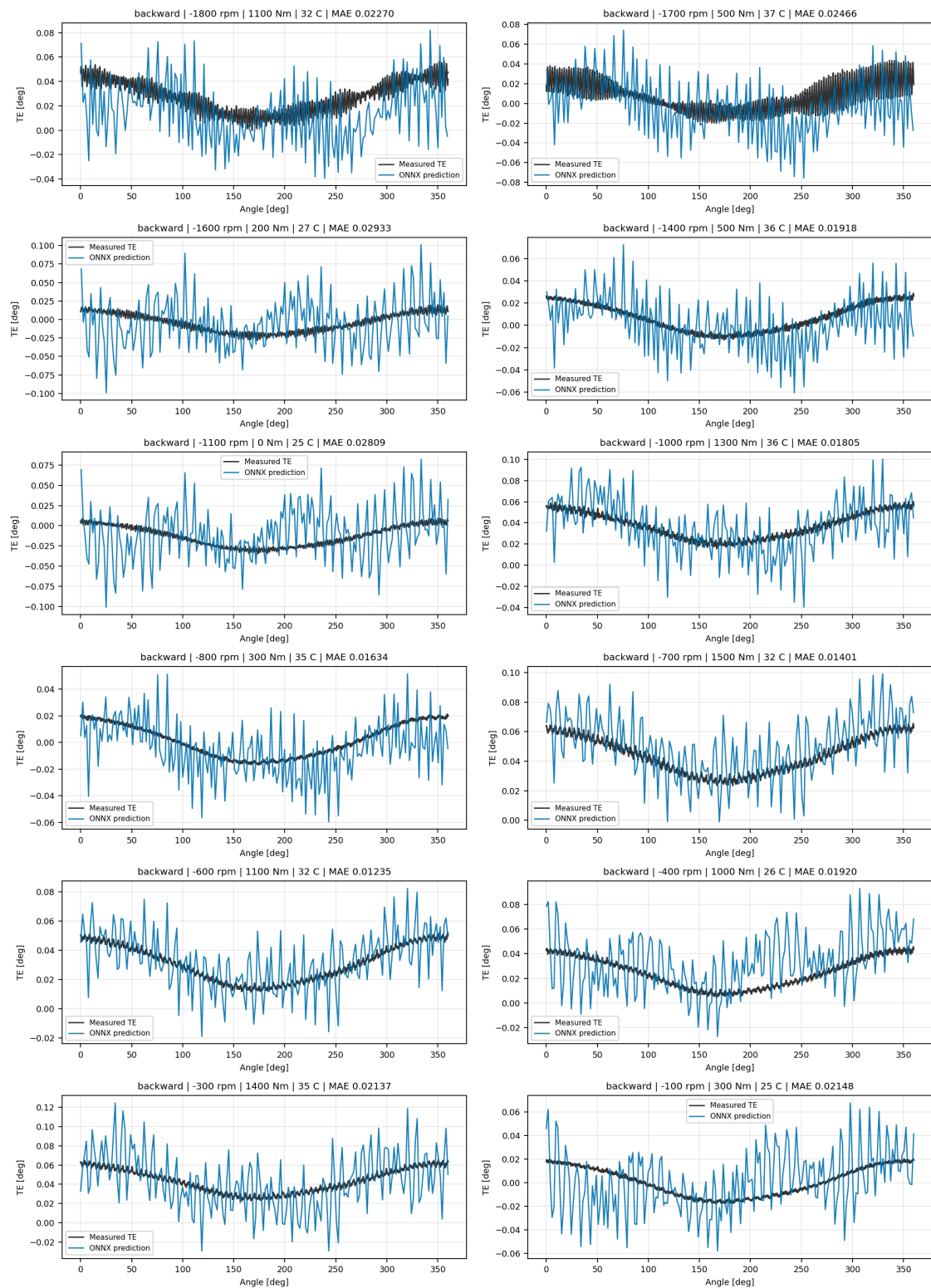
Polished Dataset + Actual Values | wave4_3_mixture_density_k2 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

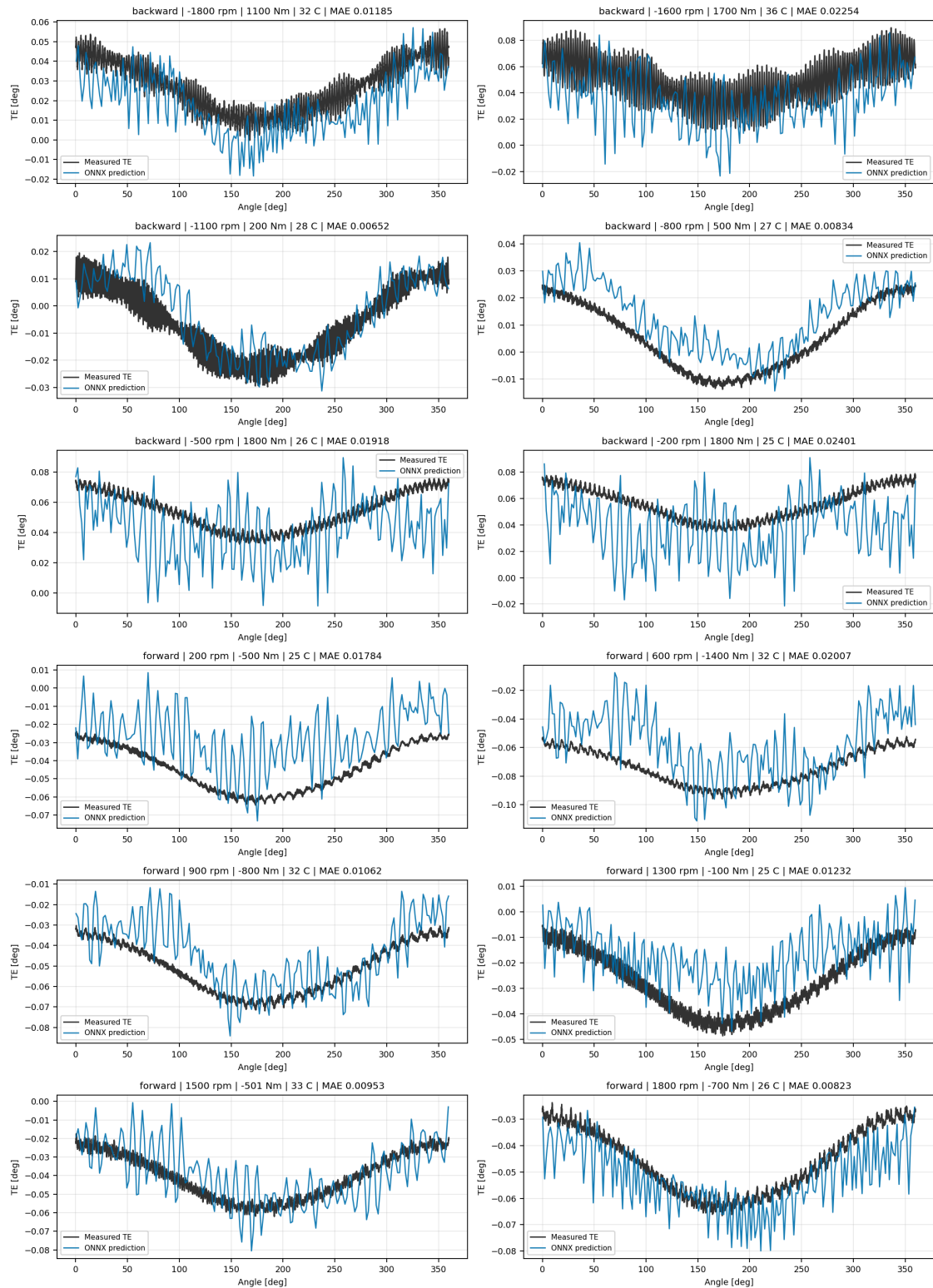
Polished Dataset + Actual Values | wave4_3_mixture_density_k2 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_3_mixture_density_k2 | global



polished_dataset_actual_values global 12-curve collage